

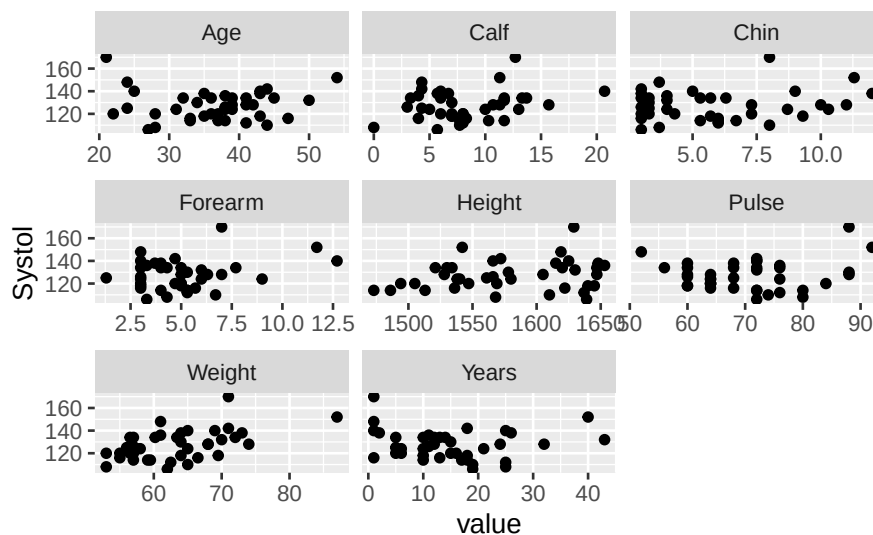
Assignment 6: Under (blood) pressure

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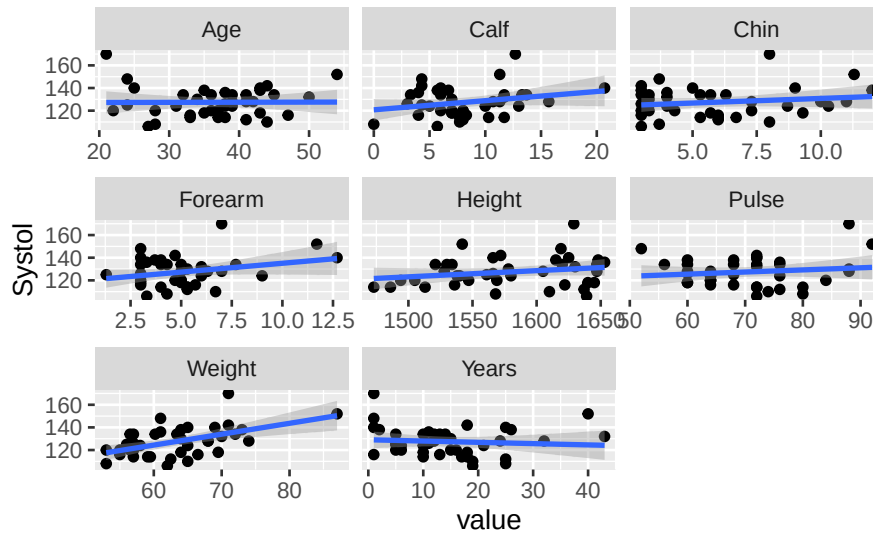
Exercise 1

```
blood_pressure %>%  
  pivot_longer(cols = Age:Pulse, names_to = "measurement", values_to = "value") %>%  
  ggplot() +  
    geom_point(mapping = aes(x = value, y = Systol)) +  
    facet_wrap(~ measurement, scales = "free_x")
```



```
blood_pressure %>%  
  pivot_longer(cols = Age:Pulse, names_to = "measurement", values_to = "value") %>%  
  ggplot() +  
    geom_point(mapping = aes(x = value, y = Systol)) +  
    facet_wrap(~ measurement, scales = "free_x") +  
    geom_smooth(mapping = aes(x = value, y = Systol), method = "lm")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Exercise 2

- What type of correlation does the Years variable have with Systol: positive, negative, or no correlation?

There is negative correlation.

- Which variable shows a moderate to strong positive correlation with Systol? (Note: the strength of the relationship increases as the slope of the trend line increases.)

Systol vs. Weight.

Exercise 3

```
blood_pressure_updated <- blood_pressure %>%
  mutate(urban_frac_life = Years / Age)
```

Exercise 4

```
systol_urban_frac_model <- lm(Systol ~ urban_frac_life, data = blood_pressure_updated)
```

Exercise 5

```
systol_urban_frac_model %>%
  tidy()
```

term	estimate	std.error	statistic	p.value
(Intercept)	133.49572	4.038011	33.059770	0.0000000
urban_frac_life	-15.75182	9.012962	-1.747686	0.0888139

```
systol_urban_frac_model %>%
  glance() %>%
  select(r.squared)
```

r.squared
0.0762564

Exercise 6

```
systol_urban_frac_df <- blood_pressure_updated %>%
  add_predictions(systol_urban_frac_model) %>%
  add_residuals(systol_urban_frac_model)
```

- What is the name of the column that holds response (y) values predicted by the model?

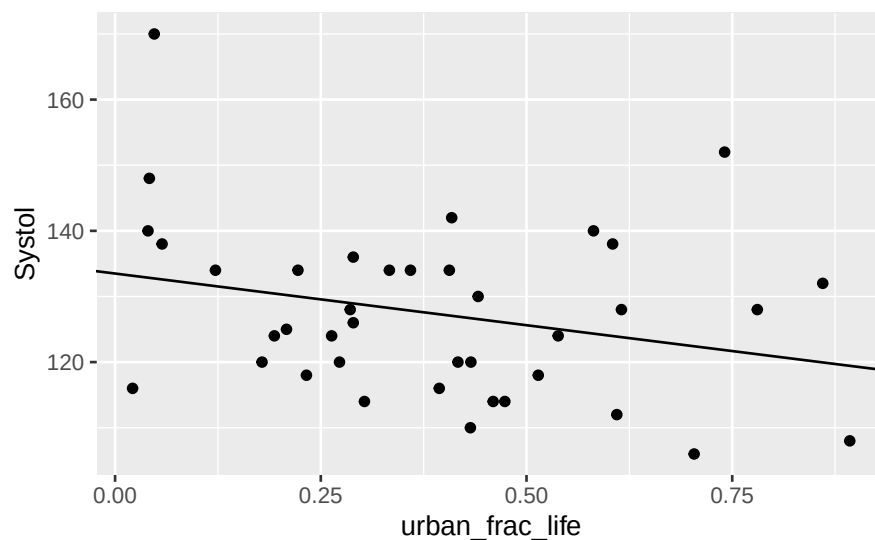
The name of the column is pred.

- What is the name of the column that holds the residuals for each observation?

The name of the column is resid.

Exercise 7

```
ggplot(systol_urban_frac_df) +
  geom_point(mapping = aes(x = urban_frac_life, y = Systol)) +
  geom_abline(slope = systol_urban_frac_model$coefficients[2],
             intercept = systol_urban_frac_model$coefficients[1])
```



- Explain whether your model meets the first condition.

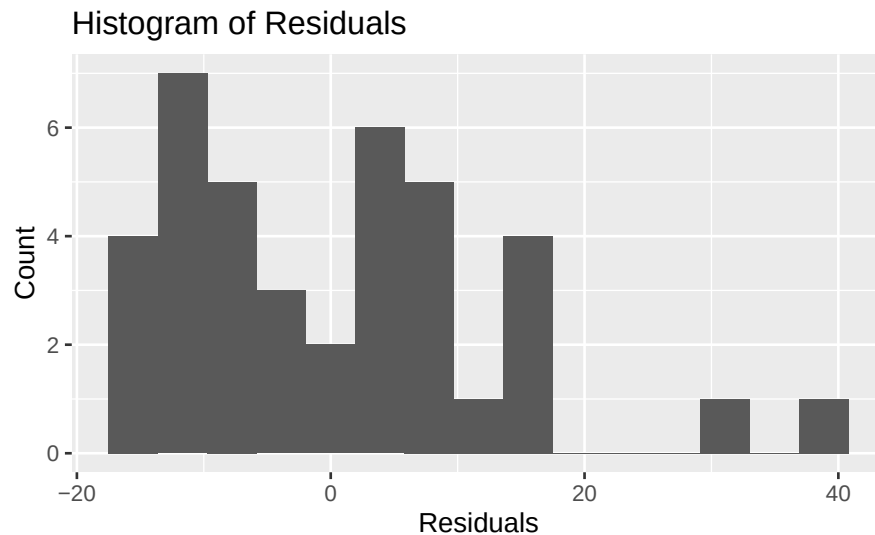
The model doesn't meet the condition. The line has a slope, this means that there is a correlation between `urban_frac_life` and `Systol`, but the points are scattered far from the line, this means that the relationship has very weak correlation.

Exercise 8

The graphs line has points scattered all around, meaning that variability isn't constant. So, the condition of constant variability isn't met.

Exercise 9

```
ggplot(systol_urban_frac_df) +  
  geom_histogram(aes(x = resid), bins=15) +  
  labs(title = "Histogram of Residuals", x = "Residuals", y = "Count")
```



- i. What is the shape of the distribution of the residuals? Symmetric? Skewed?

The shape is skewed to the left and its bimodal.

- ii. This histogram can give us a basic idea of whether the second condition, nearly normal residuals, is violated. What does your histogram suggest about whether your model meets this condition, and why?

The histogram shows that the model does not meet the nearly normal residuals condition. Since the graph is skewed to the left and has two peaks, it means the residuals don't form a normal bell shape.

Exercise 10

```
systol_weight_model <- lm(Systol ~ Weight, data = blood_pressure_updated)  
  
systol_urban_frac_model %>%  
  glance() %>%  
  select(r.squared)
```

r.squared
0.0762564

```
systol_weight_model %>%
  glance() %>%
  select(r.squared)
```

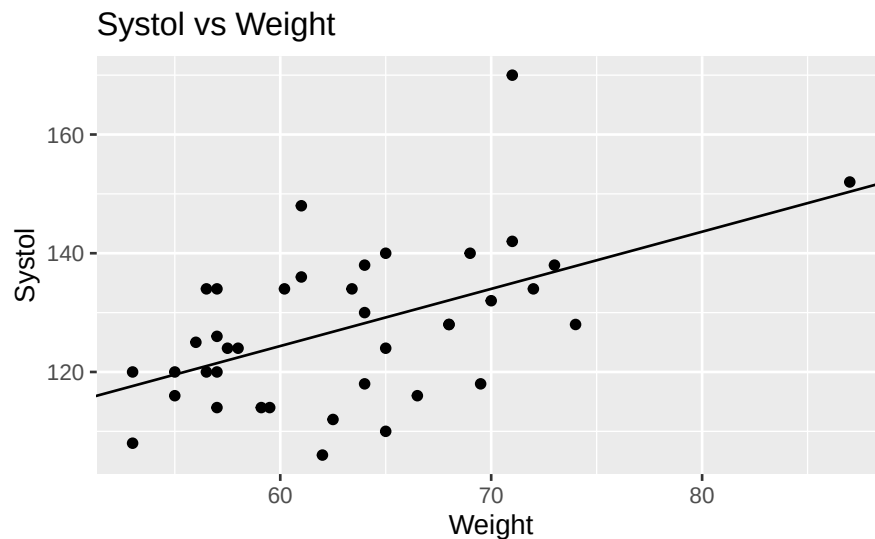
<u>r.squared</u>
0.2718207

The systol_urban_frac_model's r squared is 7% and the systol_weight_model's r squared is 27%. This means that systol_weight_model is better at predicting Systol, since it is closer to 100%.

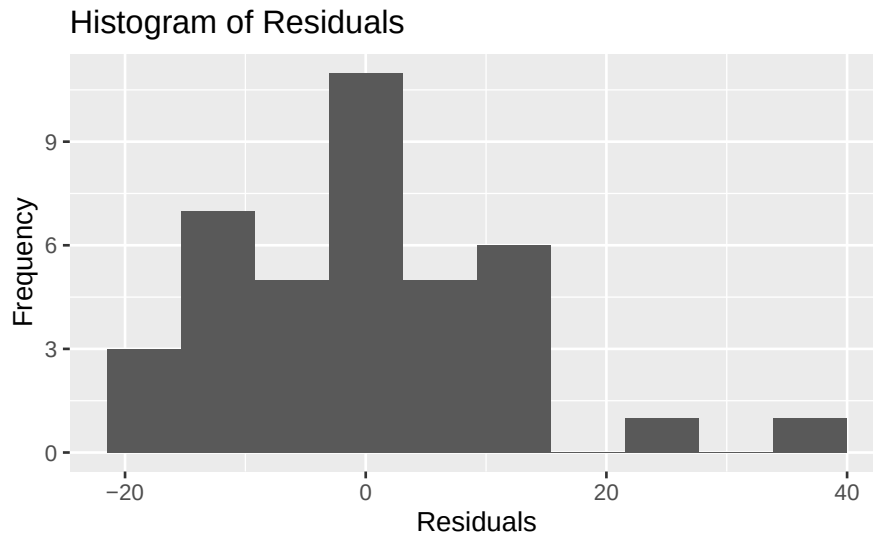
Exercise 11

```
systol_weight_df <- blood_pressure_updated %>%
  add_predictions(systol_weight_model) %>%
  add_residuals(systol_weight_model)

ggplot(systol_weight_df) +
  geom_point(mapping = aes(x = Weight, y = Systol)) +
  geom_abline(slope = systol_weight_model$coefficients[2],
              intercept = systol_weight_model$coefficients[1]) +
  labs(title = "Systol vs Weight", x = "Weight", y = "Systol")
```



```
ggplot(systol_weight_df) +
  geom_histogram(aes(x = resid), bins = 10) +
  labs(title = "Histogram of Residuals", x = "Residuals", y = "Frequency")
```



The scatter plot shows that there is a positive and weak correlation between Systol and Weight (Linearity). Also, the variability of the residuals decreases as the weight increases (Constant Variability). Finally, the histogram is more bell shaped than the last however it is still skewed to the left (Normal Residuals). This shows that the conditons aren't really met, however, it is more reliable than the last model, `systol_urban_frac_model`.

Exercise 12

Overall, the 2nd modal is more reliable than the 1st model. This is proven by the histograms, scatter plots and most importantly the r squared values, 2nd being 0.27 and 1st being 0.07.

Academic Integrity statement

No tools other than the teachers guidance from SLACK was used to complete this assignment.